DSC680 - Predicting Mental Health Treatment-Seeking Behavior in the Tech Workforce

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Project 1: Proposal
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1. Introduction

Topic: For Project 1 I am using the OSMI Mental Health in Tech Survey. I want to learn which workplace factors and attitudes line up with self-reported professional mental health treatment among people in tech. I will report patterns at the group level only—this is not meant to score employees or support individual HR decisions.

I picked OSMI because tech employers often advertise mental health benefits, but uptake still looks uneven in public discussion. The survey repeated similar questions from 2016 through 2021, so I can merge waves and wrestle with real-world mess (for example, dozens of free-text gender responses) before I model anything. I plan to train several supervised classifiers on the binary treatment indicator in the file. Because the rows are cross-sectional and self-reported, I will stick to associations in the write-up—no causal language—and use the results as a starting point for recommendations, not proof.

2. Business Problem

When mental health struggles go unsupported at work, the costs show up in turnover and lost productivity. WHO links depression and anxiety to roughly $1 trillion in global productivity losses per year (WHO, 2022). Tech gets a lot of attention for burnout, yet stigma, not knowing what benefits actually cover, and confidentiality worries still keep people from seeking care. With this data I intend to:

- Identify which workplace factors (company size, remote status, employer benefits, leave policies) most strongly predict whether someone seeks treatment
- Determine if openness about mental health at work correlates with actually getting help
- Compare multiple classification models to find the best predictor and understand which features matter most across models
- Check for demographic disparities in treatment-seeking rates and employer support

Research Questions:

1. What workplace-level factors are the strongest predictors of an employee seeking mental health treatment?

2. Does willingness to discuss mental health openly at work correlate with seeking professional treatment, even after accounting for benefit availability?

3. Which classification model (Logistic Regression, Random Forest, XGBoost, or SVM) performs best on this dataset, and what features are consistently important?

4. Are there significant demographic or geographic disparities in treatment-seeking rates within the tech workforce?

3. Datasets

My primary source is the Open Sourcing Mental Illness (OSMI) Mental Health in Tech Survey, published as anonymized CSVs on Kaggle with supporting documentation from OSMI and community maintainers (OSMI, 2021). I am relying on the dataset page for field descriptions and wave notes; there should not be personally identifying information in the public extract, which is why it is reasonable secondary data for DSC680.

1. OSMI Mental Health in Tech Survey

- Source: Kaggle distribution with original survey program at osmihelp.org; documentation describes waves and field definitions.
- Content: Self-reported job context, benefits, attitudes about discussing mental health at work, and prior use of professional mental health services.
- Format: CSV, structured tabular data
- Variables: Demographics (age, gender, country), workplace context (company size, role, remote status), employer benefits (mental health coverage, leave, EAP), attitudes (willingness to discuss, perceived stigma), and self-reported treatment history.
- Size: ~4,200 respondents across 2016–2021 survey waves
- Target Variable: treatment (binary — whether the respondent sought professional treatment)

2. BLS National Compensation Survey (Supplementary)

- Source: Bureau of Labor Statistics
- Content: Employer-provided mental health and EAP benefit rates by industry
- Purpose: Contextual benchmarking — comparing tech-sector benefit availability against national averages

4. Methods

My workflow matches what I have practiced in earlier DSC courses, applied to noisy survey inputs: harmonize the multi-year files, explore distributions and run basic chi-square checks against the outcome, engineer a small set of composites that are easy to explain, then fit and cross-validate logistic regression, Random Forest, XGBoost, and an SVM while tuning hyperparameters. For interpretation I expect to lean on SHAP visuals (Lundberg & Lee, 2017) so readers can see which features move predictions instead of only reading a table of importances.

Data Cleaning & Preparation

- Merge multi-year survey CSVs and standardize column names across waves
- Handle missing values and clean free-text fields (the gender column alone has 50+ unique entries)
- Encode categorical variables and handle any class imbalance with SMOTE

Exploratory Data Analysis

- Distribution analysis and correlation matrices for all features
- Chi-squared tests to check significance between predictors and the target
- Geographic and demographic breakdowns of treatment-seeking rates

Feature Engineering

- Create a composite "Employer Support Index" combining benefits, leave policy, EAP, and anonymity features
- Bin age into meaningful groups and apply feature selection (RFECV)

Predictive Modeling

- Logistic Regression as the interpretable baseline
- Random Forest and XGBoost for ensemble approaches
- Support Vector Machine (RBF kernel) for comparison
- Hyperparameter tuning with cross-validation
- SHAP values for model interpretability and feature importance

5. Ethical Considerations

Mental health is easy to mishandle in writing and in charts, so I am building a few guardrails up front. My plan is to:

- **Privacy Protection:** All data is anonymized with no PII. Results reported at group level only
- **Stigmatization:** Using person-first language and framing findings to advocate for support, not label individuals
- **Model Misuse Risk:** Clearly positioning the model as a policy tool, not an individual screening system that employers could use to target employees
- **Sampling Bias:** The OSMI survey skews toward English-speaking countries and male respondents — findings will be scoped accordingly
- **Informed Consent:** Survey respondents participated voluntarily knowing data would be used for research
- **Fairness:** Running demographic fairness audits to check if the model performs equitably across subgroups

6. Challenges/Issues

Anticipated challenges include:

- **Schema Inconsistencies:** Survey questions changed across years, making it difficult to unify features consistently
- **Free-Text Cleanup:** Fields like gender have dozens of variations that need careful standardization
- **Class Imbalance:** If one class dominates, accuracy becomes misleading and resampling is needed

- **Self-Report Bias:** People may not answer honestly about sensitive mental health topics
- **No Longitudinal Data:** Each respondent appears once, so no causal claims can be made — only associations
- **Generalizability:** Findings apply to tech workers specifically and may not extend to other industries

7. References

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